

Assignment 2

Experiment Tracking Notes

Working document for benchmarking and report planning

May 5, 2026

1 Purpose

This document is a working LaTeX note for tracking experiment ideas, benchmark plans, implementation decisions, and report-ready observations for the Irish-to-English speech translation assignment.

2 Confirmed Evaluation Setup

- Evaluation split: `iwslt2023_ga-eng/dev`
- Reason: `iwslt2026_ga-eng/test-2026` is a blind test set and does not contain English reference translations.
- Report requirement: clearly document the split used and the exact data source, including the linked submodule.

3 Current Dataset Notes

- Available locally:
 - audio files in `dev/wav`
 - English references in `dev/txt/dev.eng`
 - metadata in `dev/stamped.tsv`
- Important limitation:
 - no plain gold Irish transcript file was identified in this local copy
 - BLEU and chrF++ can be computed for English translation output
 - standard transcript-to-transcript ASR evaluation against gold Irish text is not directly available from these files

4 Baseline Plan

Component	Choice
ASR	<code>openai/whisper-small</code>
MT	<code>facebook/nllb-200-distilled-600M</code>
Pipeline	audio → ASR transcript → MT English translation
Metrics	BLEU, chrF++, coverage, repetition rate

5 Why A Weak Baseline Is Still Useful

- A weak baseline does not ruin the assignment.
- It provides useful material for error analysis and discussion.
- It helps motivate the improvement section.
- It lets us study error propagation from ASR to MT.

6 Planned Benchmark Experiments

6.1 Experiment 1: Baseline

- ASR: `whisper-small`
- MT: `nllb-200-distilled-600M`
- Goal: establish a standard cascaded baseline

6.2 Experiment 2: Better ASR

- Replace `whisper-small` with `whisper-medium`
- Keep the MT model fixed
- Goal: isolate the effect of better ASR quality

6.3 Experiment 3: Better MT

- Keep the baseline ASR
- Replace NLLB with another feasible GA→EN translation model
- Goal: isolate the effect of MT quality

6.4 Experiment 4: Direct Speech Translation

- Try a direct speech-to-English translation model if feasible
- Goal: compare cascaded ASR→MT against end-to-end speech translation

7 Suggested Evaluation Strategy

1. Debug each system on 50 examples.
2. Run the most stable systems on a larger subset.
3. If runtime is manageable, evaluate on the full dev split.
4. Always compare systems on the same subset in the benchmark table.

8 Metrics To Report

- BLEU
- chrF++
- Coverage
- Repetition rate

9 Error Analysis Checklist

- ASR substitution
- ASR deletion
- ASR hallucination
- MT untranslated tokens
- MT mistranslation
- MT over-generation
- Error propagation from ASR into MT

10 Implementation Notes

10.1 Notebook Variables

Listing 1: Current dataset path setup

```
DATA_ROOT = Path("iwslt2023_ga-eng")
DEV_ROOT = DATA_ROOT / "dev"
AUDIO_DIR = DEV_ROOT / "wav"
TRANS_FILE = DEV_ROOT / "stamped.tsv"
REF_FILE = DEV_ROOT / "txt" / "dev.eng"
```

10.2 Current ASR Note

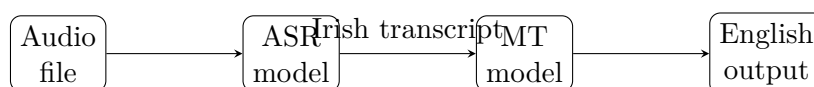
Whisper was tested as the baseline ASR model. In this setup, explicit Irish forcing with `language='ga'` was not available for the chosen Whisper model variant, so this limitation should be documented in the report.

11 Benchmark Table Template

System	BLEU	chrF++	Coverage	Rep. Rate
Baseline: Whisper-small + NLLB-600M	–	–	–	–
Improved ASR: Whisper-medium + NLLB-600M	–	–	–	–
Improved MT: Whisper-small + Alternative MT	–	–	–	–
Direct ST model	–	–	–	–

12 Diagram Area

This section can be expanded later with architecture diagrams or workflow figures.



13 Code Snippet Area

Use this area later to store code excerpts worth reusing in the report.

Listing 2: Placeholder for future code excerpt

```
def placeholder():  
    pass
```

14 Report Framing Ideas

- Improving ASR should reduce downstream translation errors.
- Better MT cannot fully recover from a poor transcript.
- Direct speech translation may avoid some cascade errors, but may also be less robust depending on the model and language coverage.
- Any limitation around Irish language support must be reported clearly.

15 Next Steps

1. Finalize the baseline pipeline.
2. Compute baseline metrics.
3. Inspect outputs manually.
4. Compare at least one improved ASR system.
5. Compare at least one improved MT system if feasible.
6. If time allows, try a direct speech translation model.